

Detecting AI-Generated Chinese Social Media Posts Using Linguistic and Semantic NLP Features

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Abstract

Generative AI has made AI-written content increasingly common on Chinese social media platforms such as Xiaohongshu and Weibo, where posts can influence public opinion, product perception, and user trust. However, detecting AI-generated Chinese text remains challenging due to informal language styles, slang, emojis, and code-switching commonly used in online communication. In addition, most existing AI-text detection studies focus on English datasets and formal writing, leaving short-form Chinese social media content relatively under-explored. This proposal presents an NLP-based framework for distinguishing AI-generated Chinese social media posts from human-written content using linguistic and semantic features. The study will compare traditional machine learning methods, transformer-based models, and LLM-as-classifier approaches to evaluate their effectiveness in AI-text detection.

1 Introduction

1.1 Problem Statement

Current AI-generated text detection methods face several limitations when applied to Chinese social media platforms. Existing detectors are mostly trained on English datasets and struggle to generalize to informal Chinese online language. Furthermore, transformer-based language models continue to improve, making AI-generated content increasingly difficult to distinguish from authentic human-written posts.

As a result, there is a growing need for robust NLP-based methods capable of detecting AI-generated Chinese social media content.

1.2 Research Objectives

The objectives of this research are as follows:

- To analyze linguistic and semantic differences between AI-generated and human-written Chinese social media posts.

- To construct a labeled dataset containing authentic human-written content and AI-generated text.
- To compare the effectiveness of traditional machine learning models and transformer-based NLP models for AI-generated text detection.
- To evaluate the contribution of stylistic and semantic features in improving detection performance.

1.3 Hypothesis

We hypothesize that transformer- and LLM-based detectors will outperform TF-IDF classifiers by better capturing contextual and semantic patterns in Chinese social media posts.

2 Related Work

Recent studies have explored AI-generated Chinese text detection using both transformer-based and large language model approaches. [Wang et al. \(2024\)](#) showed that instruction-tuned LLMs achieved stronger cross-domain robustness than traditional BERT- and RoBERTa-based detectors. Similarly, [Su et al. \(2025\)](#) found that AI-generated Chinese text often exhibits repetitive lexical patterns and lower vocabulary diversity. More recently, [Qing et al. \(2026\)](#) introduced C-ReD, a large-scale Chinese benchmark covering multiple writing domains and LLM generators, highlighting that detection difficulty varies significantly across domains. Despite recent progress, AI-generated short-form Chinese social media content remains relatively underexplored.

3 Methodology

3.1 Dataset Collection

This study uses the RedNote-Vibe dataset, which is based on Xiaohongshu/RedNote content and focuses on informal Chinese social media writing.

The dataset contains human-written, AI-generated, and exploration subsets. Only the human-written and AI-generated training subsets will be used for binary classification, while the exploration subset will be excluded because it mainly supports temporal analysis.

Compared with general Chinese AI-text datasets, RedNote-Vibe better reflects real social media environments through its use of informal expressions, emojis, hashtags, and platform-specific writing styles.

3.2 Data Preprocessing

The selected JSONL files will be converted into a structured format containing post text and labels, then shuffled and split into training, validation, and test sets.

Preprocessing steps include removing duplicate or empty posts, punctuation normalization, whitespace cleaning, and handling emojis, hashtags, and user mentions through placeholder normalization. Chinese word segmentation will be performed using *jieba*. Stop-word removal may be applied for TF-IDF features, while transformer-based models will use lightly cleaned text to preserve contextual information.

The processed dataset will support both linguistic and semantic feature extraction, including TF-IDF n-grams, vocabulary diversity, repetition patterns, emoji usage, and contextual embeddings from pretrained Chinese transformer models.

3.3 Feature Extraction

This study extracts both linguistic and semantic features from Chinese social media posts in the RedNote-Vibe dataset.

Linguistic Features The linguistic features include post length, sentence length, vocabulary diversity, emoji frequency, punctuation usage, hashtag frequency, and repetition patterns. In addition, TF-IDF character-level and word-level n-gram features will be extracted to capture common lexical and stylistic patterns in AI-generated text.

Semantic Features Semantic features will be extracted using pretrained Chinese transformer models such as Chinese BERT and RoBERTa. These features include contextual embeddings, semantic coherence between sentences, sentiment polarity, and sentiment consistency across each post. Such features aim to capture deeper semantic and con-

textual differences between human-written and AI-generated content.

3.4 Model Development

This study will compare three different models: 1. TF-IDF + SVM, 2. Chinese RoBERTa, and 3. LLM-as-Classifier.

3.5 Evaluation Metrics

Model performance will be evaluated using accuracy, precision, recall, and AUROC.

4 Expected Results

This research is expected to:

- Identify effective linguistic and semantic features for detecting AI-generated Chinese text.
- Compare the effectiveness of traditional and transformer-based NLP approaches.
- Contribute toward Chinese NLP research on AI-generated content detection.
- Propose a practical framework for improving authenticity verification on social media platforms.

5 Timeline

Phase	Duration
Literature Review	Week 12–13
Dataset Collection	Week 12–13
Data Preprocessing	Week 12–13
Model Development	Week 13–14
Evaluation	Week 13–14
Project Presentation	Week 15
Report Submission	Week 15

Table 1: Proposed research timeline

References

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